0.1 Chemical composition and molecular-size coverage

We compared the chemical content of CPOSS with the training split of Open Molecular Crystals 2025 (OMC25)[1], counting each documented CPOSS molecular system and each OMC25 CSD reference code once (Fig. 1). The principal comparison contains 190 single-component CPOSS systems with coordinate-validated compositions and 44,403 OMC25 molecular records. Repeated CSP searches, energy models, generated packings and relaxation frames do not weight these molecular distributions. A further 44 CPOSS multicomponent systems are described separately on their complete stoichiometric formula-unit basis. One CPOSS system with no valid alternative coordinate element table was excluded from the coordinate-validated population; including the reported formula gives a supplementary metadata-only population of 191 single-component systems.

CPOSS single-component systems contain 5–100 atoms per molecule (median 18), compared with 4–164 atoms (median 39) in OMC25. The corresponding median non-hydrogen atom counts are 12 and 21. Using one average-atomic-weight convention for both datasets, CPOSS molar masses span 32.04–890.88 $\mathrm{g\,mol^{-1}}$ (median 179.69 $\mathrm{g\,mol^{-1}}$), whereas OMC25 spans 30.03–1386.32 $\mathrm{g\,mol^{-1}}$ (median 312.15 $\mathrm{g\,mol^{-1}}$). These summaries describe the observed molecular populations and do not establish equivalence of their chemical environments or crystal packings.

Elemental occurrence is expressed as the fraction of molecular records containing each species. Oxygen occurs in 78.9% of CPOSS systems and 81.9% of OMC25 molecular records, while nitrogen occurs in 71.6% and 65.4%, respectively. The verified CPOSS population also contains Cu, outside the elemental vocabulary of OMC25. Its occurrence does not imply that OMC25 models have been evaluated on this chemistry. CPOSS additionally includes salts, solvates and co-crystals, which require a distinct formula-unit basis and are outside OMC25's stated single-component scope. The comparison therefore describes complementary sampling priorities: repeated crystal-energy-landscape searches in CPOSS and broad molecular sampling for training interatomic potentials in OMC25. The supplied reference figure provides no CSD elemental, molecular-size or mass distributions, so no such CSD curves are inferred here.

References

[1] Vahe Gharakhanyan, Luis Barroso-Luque, Yi Yang, Muhammed Shuaibi, Kyle Michel, Daniel S. Levine, Misko Dzamba, Xiang Fu, Meng Gao, Xingyu Liu, et al. Open Molecular Crystals 2025 (OMC25) dataset and models. *Scientific Data*, 2026. doi: 10.1038/s41597-026-06628-2. URL `https://www.nature.com/articles/s41597-026-06628-2`.

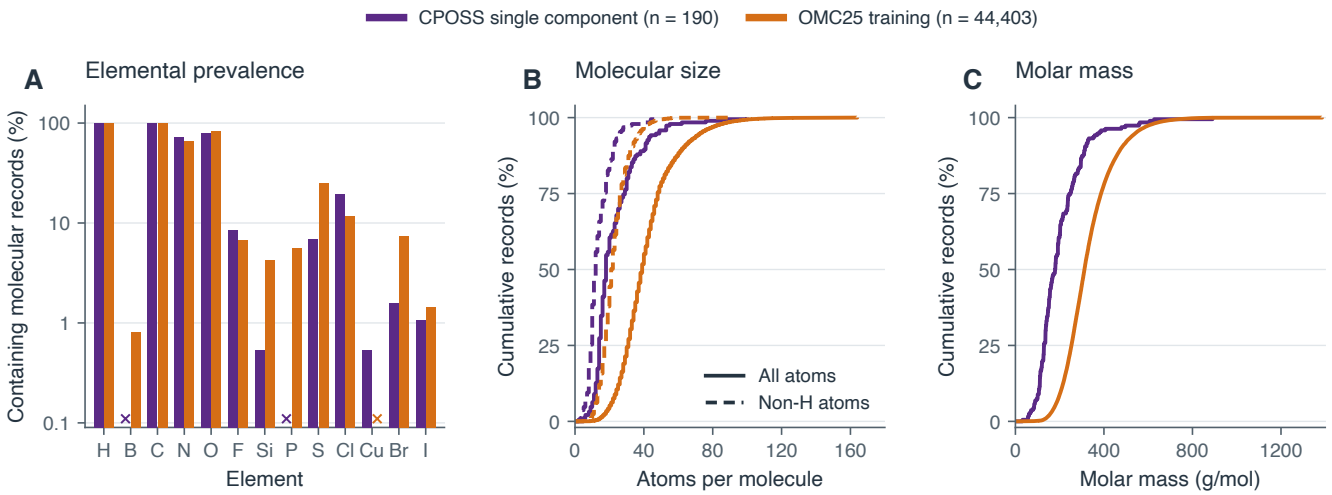


Figure 1: **Chemical content of CPOSS and OMC25.** The comparison uses 190 single-component CPOSS molecular systems with compositions checked against archived coordinate representatives and 44,403 molecular records from the OMC25 training split. Each CPOSS system is counted once by its documented source page, and each OMC25 molecule once by its CSD reference code after verifying invariant molecular metadata across its starting-crystal records. Records are not merged by molecular formula. Repeated searches, energy models, crystal packings and relaxation frames do not weight the distributions. **(A)** Percentage of molecular records containing each element, using the union of species present in either dataset. Categories overlap and need not sum to 100%. The ordinate is logarithmic; crosses at the lower edge denote zero occurrence. **(B)** Empirical cumulative distributions of molecular atom counts, including all hydrogens (solid) or excluding hydrogen (dashed). **(C)** Empirical cumulative distributions of molar mass, calculated from complete molecular compositions using the same frozen RDKit average atomic weights for both datasets. Official OMC25 `mol.mass` values are retained in the data tables; harmonisation changes individual training-set masses by at most $0.032\,\mathrm{g\,mol^{-1}}$. All ECDFs are evaluated without binning. The source audit's eight composition conflicts and two malformed website formulas were reviewed against workbook and coordinate evidence, with all source choices recorded. One CPOSS system with no valid alternative coordinate element table was excluded from the primary comparison and retained in a separate metadata-only sensitivity analysis. The 44 verified multicomponent CPOSS systems are shown separately in the supplementary comparison using complete written stoichiometry or explicitly documented coordinate-derived component ratios. OMC25's molecular basis follows its published single-component definition; connectivity was not independently reconstructed from its metadata CSV. These molecular distributions do not describe unit-cell sizes or counts of independent crystal minima.

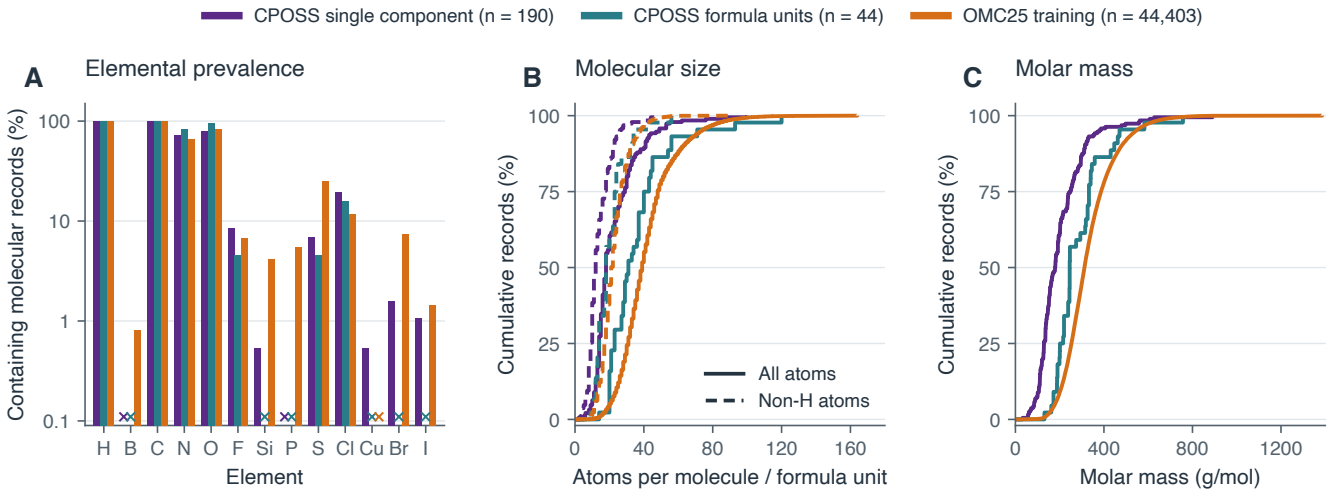


Figure 2: **Single-component and multicomponent CPOSS coverage relative to OMC25.** The three populations are 190 coordinate-validated single-component CPOSS systems, 44 coordinate-validated multicomponent CPOSS formula units, and 44,403 OMC25 training molecular records. Panels and graphical conventions follow Fig. 1. Multicomponent systems include salts, hydrates, solvates and co-crystals and retain their full stoichiometric compositions, including counterions, coformers and water. Their atom counts and molar masses describe a formula unit rather than an individual molecule; they are shown as a distinct descriptive population. The complete written stoichiometry is preserved where it agrees with the archived component counts. For QuinoliniumFumarate and BTXV, coordinate-derived minimum integer component ratios resolve conflicting source compositions. Percentages and ECDFs are normalized independently within each population and are not weighted by the number of molecular components, CSP searches or archived structures.